llmXive Automated Review of Ideas Have Genomes: Benchmarking Scientific Lineage Reasoning and Lineage-Grounded Idea Generation

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the llmXive agent-prompt library; each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

The paper presents a novel benchmark for scientific lineage reasoning, but several factual claims regarding the evaluated systems and data composition require verification or clarification to ensure accuracy.

The most critical issue concerns the evaluation baselines. The text and Table 1 repeatedly cite "GPT-5.5" and "Claude Opus 4.7" as the primary models tested. As of the current date, neither of these models has been released by OpenAI or Anthropic. The reported performance metrics (e.g., 27.3% exact accuracy for GPT-5.5) are therefore unverifiable and potentially hallucinated or based on hypothetical future models. If these are placeholders for existing models (e.g., GPT-4o or Opus 3.5), the names must be corrected to reflect the actual systems used. If the paper is a projection or simulation, this must be explicitly stated, as presenting them as empirical results against non-existent models invalidates the experimental claims.

Additionally, while the numbers for the benchmark construction (1,961 traces, 1,085 objects) are consistent across the abstract and Section 4.1, the relationship between the 1,961 "golden lineage traces" and the 1,029 "instances" in IG-Exam is not explicitly defined. Readers may assume a direct 1:1 mapping or a simple subset relationship. Clarifying how the 1,029 instances were derived from the 1,961 traces (e.g., multiple instances per trace, or a specific subset) would improve the precision of the data description.

Finally, the phrasing in Section 5.2 regarding "GPT-5.5 + Claude Code" reaching 27.3% is slightly ambiguous compared to the table label "Claude Code (GPT-5.5)". While likely referring to the same row, precise alignment between the narrative description and the table labels is necessary to avoid confusion about the specific system configuration.

These issues are primarily fixable by correcting the model names to reflect reality and clarifying the data derivation, but the presence of non-existent models in the main results table is a significant accuracy concern that must be resolved.

Action items.

- [**writing**] The paper presents a novel benchmark for scientific lineage reasoning, but several factual claims regarding the evaluated systems and data composition require verification or clarification to ensure accuracy. The most critical issue concerns the evaluation baselines. The text and Table 1 repeatedly cite "GPT-5.5" and "Claude Opus 4.7" as the primary models tested. As of the current date, neither of these models has been released by OpenAI or Anthropic. The reported performance metrics (e.g., 27.

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_figure_critic.md`

Figure 1

The heatmap effectively visualizes the data with clear numerical values and a color scale, but the x-axis domain abbreviations are undefined and the y-axis labeling style is inconsistent.

Figure 2

The heatmap effectively displays the data with clear numerical values and a color scale, but the caption fails to define the H/V/S abbreviations or explain the color coding used for the model names on the y-axis.

Figure 3

The stacked bar chart is generally clear but contains a discrepancy in the 'Question' bar where the legend-defined 'Niche Competition' category is missing from the visual stack, and the visible segments do not sum to 100%.

Figure 4

The figure effectively visualizes the H/V/S metrics but suffers from poor x-axis label legibility due to rotation and overlap. Additionally, the sorting order of the systems does not strictly align with the caption's claim of being sorted by aggregate Lineage PES.

Figure 5

Figure 5 presents a heatmap and bar chart but has inconsistencies between the heatmap data and bar chart values, and lacks explicit system definitions and colorbar units in the caption.

Figure 6

The Sankey diagram effectively visualizes the volume of errors flowing from tiers to fields to classes, but the legend is illegible and fails to map the specific error types listed to the E1-E9 categories, hindering the interpretation of specific failure modes.

Figure 7

Figure 7 is a clear and well-structured schematic that effectively visualizes the evaluation design described in the caption. The pipeline steps, input types, and distinct modules for 'IdeaGene-Exam' and 'IdeaGene-Arena' are legible and logically organized, successfully supporting the claim of converting papers into a lineage substrate for dual capability evaluation.

Figure 8

Figure 8 is a clear, well-structured conceptual diagram that effectively contrasts paper-centric and genome-centric views of scientific ideas. The visual metaphors (monsters, tree, flowcharts) are intuitive, and the caption accurately describes the transition from unstructured paper stacks to explicit lineage structures. All components are legible, and the evolutionary mechanisms (Mutation, Radiation, etc.) are clearly labeled and illustrated.

Figure 9

The figure effectively visualizes PES gains across settings, but the caption is overly generic and the axis label does not clarify that the plotted values represent gains or deltas rather than absolute scores.

Figure 10

The radar chart in Figure 10 is visually clear with a complete legend, but the caption is overly generic and the radial axis scale lacks a defined minimum value.

Action items.

- [**writing**] Figure 1: The x-axis labels (e.g., 'CS', 'Neuro') are abbreviations that are not defined in the caption or on the axis itself, making the specific domains ambiguous to the reader.
- [**writing**] Figure 1: The y-axis labels use inconsistent formatting; some entries are prefixed with '+' (e.g., '+ Claude Code') while others are not, and the color coding for these prefixes is not explained in the caption.
- [**writing**] Figure 2: The caption 'PES rubric decomposition (H / V / S)' does not define the abbreviations H, V, and S; the x-axis labels ('Heredity', 'Variation', 'Selection') should be explicitly linked to these abbreviations in the caption for clarity.
- [**writing**] Figure 2: The y-axis labels use inconsistent color coding (blue, orange, green) without a legend or caption explanation to define what these colors signify (e.g., model families or categories).
- [**science**] Figure 3: The legend defines 'Niche Competition' (dark grey), but this color is not visible in the 'Question' bar despite the bar reaching 100%. The visible segments (Mutation, Adaptive Radiation, Hybridization, Speciation) sum to ~98%, leaving a gap that is not accounted for by the 'Other' (light grey) category or the missing 'Niche Competition'

segment.

- [**writing**] Figure 3: The legend entry for 'Other' (light grey) does not appear to correspond to any visible segment in the 'Question' bar, creating ambiguity about whether the missing ~2% is 'Other' or 'Niche Competition'.
- [**science**] Figure 4: The x-axis labels are rotated 45 degrees and overlap significantly (e.g., 'G5.5 + Codex' vs 'G5.5 + Claude'), making the system names difficult to read and distinguish.
- [**science**] Figure 4: The caption claims systems are 'sorted by aggregate Lineage PES', but the visual ordering of the bars does not appear to follow a strict descending or ascending order of the average height of the three bars (e.g., 'G5.5 + Codex' is first, but 'G5.5 + Claude' has a higher average).
- [**writing**] Figure 4: The y-axis label 'Subscore' is generic; the caption defines the metrics as Heredity, Variation, and Selection, but the axis does not specify the unit or scale range (e.g., 0-100).
- [**science**] Figure 5: The heatmap contains 14 rows of data, but the caption states 'all evaluated systems' without listing them; the row labels (e.g., 'GPT-5.5 + Claude Code') are not defined in the caption or cross-referenced captions, making it impossible to verify if the systems match the paper's scope.
- [**writing**] Figure 5: The colorbar legend is present but lacks a title or unit label (e.g., 'Exact accuracy (%)'), which is only implied by the y-axis label of the adjacent bar chart; this should be explicitly stated for clarity.
- [**science**] Figure 5: The bar chart on the right shows 'Axis mean' values (e.g., T1=28.5) that do not match the average of the corresponding column in the heatmap (e.g., T1 column average 28.1), suggesting a calculation or labeling inconsistency.
- [**science**] Figure 6: The legend at the top lists specific error types (e.g., 'verify', 'swapped_gene_role') that are not mapped to the E1-E9 error classes shown on the right axis, making it impossible to trace specific error mechanisms as the caption implies.
- [**writing**] Figure 6: The legend at the top is rendered in a font size that is illegible and overlaps with the Sankey flows, rendering the specific error type labels unreadable.
- [**writing**] Figure 9: The caption 'PES across information settings' is too generic and fails to describe the specific comparison (Question vs. Library vs. Lineage) or the metric (PES gain) shown in the plot.
- [**science**] Figure 9: The y-axis labels are color-coded (blue, green, orange) to match the legend, but the text colors for 'G5.5 + Codex' and 'G5.5 + Claude' are orange, while the corresponding data points are green and orange respectively, creating potential confusion between model names and information settings.
- [**writing**] Figure 9: The x-axis label 'PES (0-100)' is present, but the plot displays delta values (e.g., +4.4, +9.9) next to the points, which are not explicitly defined in the axis label or caption as 'PES Gain' or 'Delta PES'.

- [**writing**] Figure 10: The caption ‘Five capability dimensions’ is too vague; it should explicitly list the five dimensions (e.g., Inheritance Tracing, Evolutionary Reasoning, etc.) shown on the radar chart axes.
- [**writing**] Figure 10: The radial axis label ‘best’ is ambiguous without a corresponding ‘worst’ or ‘0’ label at the center, making the scale direction implicit rather than explicit.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_jargon_police.md`

The manuscript demonstrates excellent self-containment for a competent reader from an adjacent field (e.g., a researcher in NLP, AI safety, or computational social science). The authors successfully avoid the common pitfall of assuming familiarity with their specific subfield's private vocabulary.

Key strengths in accessibility include:

1. **Operational Definitions:** The core concepts—`Idea Genome` (genome), `GenomeDiff` (genomediff), and the six `Evolutionary Dynamics`—are rigorously defined upon first introduction in Section 2 (Framework) and Section 3 (Benchmark). The paper explicitly states that these are “operational categories” rather than requiring a broad biological ontology, which immediately grounds the reader.
2. **Acronym Management:** All custom acronyms (IG-Bench, IG-Exam, IG-Arena, PES) are expanded at their first occurrence in the Abstract or Introduction. Standard field terms (LLM, RL, Transformer, ELO) are used correctly without needing definition, as they are foundational to the target audience.
3. **Notation Clarity:** Mathematical notation (e.g., Equation 1 for the genome object, Equation 2 for PES) is accompanied by clear prose explanations of every variable ($t_i, z_i, e_i, c_i, H, V, S$). There are no overloaded symbols or undefined operators.
4. **Contextualization:** When referencing specific benchmarks or methods (e.g., SWE-bench, GAIA, Chatbot Arena), the authors provide brief, one-sentence glosses explaining their relevance or nature, ensuring a reader outside the immediate LLM-evaluation niche can follow the comparison.

The paper avoids “in-group shorthand” and buzzwords by consistently linking abstract terms (like “lineage competence”) to concrete, measurable tasks (T1-T4). A reader from a neighboring discipline could follow the logic of the benchmark construction and the interpretation of the results without needing to consult external literature to decode the terminology. No undefined terms or opaque symbols were found.

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_logical_consistency.md`

The paper's argument structure is logically sound and internally consistent. The definitions of the core concepts—specifically the `Idea Genome` (Section 2.2), `GenomeDiff` (Section 2.4), and the six `Evolutionary Dynamics` (Section 2.5)—are established clearly and applied consistently throughout the benchmark construction (Section 3) and experimental evaluation (Section 4).

The logical flow from the problem statement (paper-centric retrieval is insufficient for lineage reasoning) to the proposed solution (genome-centric representation) to the empirical validation is coherent. The distinction between the two evaluation modes, `IG-Exam` (closed-form reasoning) and `IG-Arena` (open-ended generation), is maintained without contradiction. The metrics defined (Exact Accuracy for Exam, PES for Arena) align perfectly with the stated goals of each section.

Specifically, the causal claims regarding the "plausibility-coherence gap" (Finding 4) are supported by the decomposition of the PES metric into Heredity, Variation, and Selection, as presented in the results. The paper correctly attributes the performance gap to Heredity scores rather than Variation, which is a valid inference from the provided data breakdown. The limitations section appropriately qualifies the scope of the evolutionary dynamics without contradicting the main results. No non-sequiturs, circular arguments, or numerical inconsistencies were found.

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_overreach.md`

The paper generally maintains a strong alignment between its evidence and its claims, particularly in the careful definition of the "Idea Genome" and the specific metrics used. However, there are three instances where the rhetoric slightly exceeds the scope of the demonstrated evidence, primarily regarding the generalizability of the model pool and the interpretation of correlation as design prescription.

First, the Abstract and Introduction frame the evaluation as "Experiments on 14 LLM-based scientists," which suggests a broad, diverse survey of the current state of the art. However, the results in Section 5.1 and Table 1 reveal that the participant pool is heavily concentrated on a single backbone: 12 of the 14 entries are variants of GPT-5.5 (direct, agents, or CLI harnesses), with only two using Claude Opus 4.7. No open-weight models (e.g., Llama, Mistral) or other major closed models are represented in the main comparison. The claim of "14 scientists" creates an impression of a wide-ranging benchmark that the specific model selection does not fully support. The text should be narrowed to reflect that the study evaluates "14 configurations primarily based on GPT-5.5 and Claude Opus" to avoid implying a generalizable finding across all model families.

Second, the Abstract and Conclusion state that structured lineage context "reshuffles system rankings rather than helping every participant uniformly." While the data in Section 5.3 does show heterogeneous gains (e.g., GPT-5.5 gains +2.3 while Kimi gains +6.9), the phrasing "rather than helping" subtly implies the context might be ineffective or neutral on average. The text explicitly notes a "median gain is +4.4," confirming that the context *does* help on average. The rhetoric should be adjusted to "reshuffles rankings despite providing an average gain" to ensure the reader understands the context is beneficial overall, even if the distribution of that benefit is uneven.

Finally, the Conclusion asserts that the findings "point to a concrete design direction: auto-research systems need compositional verification modules." This is a prescriptive claim derived from the moderate positive correlation shown in Figure 5.4b between closed-form understanding and generation heredity. While the correlation supports the hypothesis, the experimental design (observational correlation) does not prove that adding verification modules *causes* better generation, nor does it rule out that both capabilities stem from a third factor (e.g., general reasoning ability). The language should be hedged to "suggest that compositional verification modules may be a necessary component" or "indicate a potential design direction" to align the prescriptive strength with the correlational evidence.

Action items.

- [**writing**] Abstract: 'Experiments on 14 LLM-based scientists' implies broad diversity, but Section 5.1 shows 12/14 are GPT-5.5 variants. Narrow to '14 configurations primarily based on GPT-5.5' or explicitly note the lack of open-weight model testing.
- [**writing**] Abstract/Conclusion: 'Reshuffles rankings rather than helping' implies ineffectiveness, yet Section 5.3 notes a median gain of +4.4. Rephrase to 'reshuffles rankings despite providing an average gain' to accurately reflect the data.
- [**writing**] Conclusion: 'Need compositional verification modules' prescribes a causal solution from a correlational finding (Fig 5.4b). Change to 'suggest that compositional verification modules may be a necessary component' to align with the evidence.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_safety_ethics.md`

This paper presents a benchmark for evaluating scientific lineage reasoning and idea generation. The work is low-risk by construction. The dataset consists of "golden lineage traces" and "curated genome objects" derived from published scientific literature (e.g., YOLO, BERT, DETR) and frontier questions. The data construction process explicitly mentions “anonymization leakage” checks in Section 4.1, and the examples provided in the Appendix (e.g., Appendix A.2) use anonymized or generic descriptions of scientific concepts rather than raw, sensitive personal data.

The “human subjects” component involves 50 graduate annotators validating the benchmark construction and difficulty. The paper states in the Appendix (“Human Agreement”) that these annotators were recruited to validate labels and solve items. While the paper does not explicitly cite an IRB approval number or exemption statement, the nature of the task (validating benchmark items and solving reasoning problems on public scientific papers) typically falls under exempt categories in many jurisdictions, or involves minimal risk where formal IRB review is not strictly mandated by all venues, though it is best practice. However, given the context of benchmark construction using public data and expert annotators performing standard validation tasks, the absence of a specific IRB citation is not a fatal safety failure, nor does it indicate a foreseeable, non-trivial risk of harm that is unmitigated. The risk of re-identification or privacy violation is negligible as the data is derived from public publications and the annotators are consenting experts.

There are no dual-use capabilities described that lower the barrier to harm (e.g., automated vulnerability discovery, biological synthesis). The “idea generation” capability is evaluated on scientific reasoning, not on generating harmful content, disinformation, or deceptive personas. The paper does not release any PII, nor does it scrape data in violation of terms of service (it uses existing public papers and curated traces).

Consequently, there are no specific, nameable gaps in disclosure or mitigation that require action. The paper is safe to publish as is.

Scientific Evidence

Weighs the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_scientific_evidence.md`

The paper presents a novel benchmark for scientific lineage reasoning, but the evidentiary strength of the experimental results is currently insufficient to support the specific claims of system differentiation and ranking stability.

The primary concern is the complete absence of variance reporting in the main results. Table 1 presents exact accuracy scores for 14 different systems on the \geneexam{} benchmark (1,029 instances) as single point estimates (e.g., 27.3% for Claude Code vs 23.1% for GPT-5.5). In LLM evaluation, performance on a fixed test set can fluctuate significantly based on random seed, temperature, or prompt phrasing. A difference of 1-2 percentage points is well within the margin of error for a single run on a dataset of this size. Without reporting standard deviations across multiple seeds or confidence intervals, the claim that “structured lineage context reshuffles system rankings” (Section 5.3) is not statistically grounded. The observed ranking changes could easily be artifacts of a lucky seed for one system and an unlucky one for another, rather than a genuine effect of the lineage context.

Similarly, the \genearena{} results rely on a Population-Evolution Score (PES) derived from a panel of three model judges. The paper reports Krippendorff's α for agreement but does not report the variance of the PES scores themselves across different judge seeds or runs. If the judge panel's scoring is noisy, the small differences in reported PES (e.g., 86.5 vs 86.7) are uninterpretable. The claim that "Variation and Selection stay nearly constant... while Heredity explains the gap" requires evidence that these sub-scores are stable across runs, not just that they are constant in a single snapshot.

To close this gap, the authors must report results across at least 3-5 random seeds for both the \geneexam{} and \genearena{} evaluations. This should include mean $\pm$ standard deviation for all accuracy and PES metrics. Additionally, the stability of the system rankings should be verified; if the ranking changes significantly across seeds, the claim of a "reshuffle" is not robust. Finally, the variance of the judge panel's scores should be reported to ensure that the PES metric is distinguishing signal from noise. Without these additions, the reported "compositional bottleneck" and specific system comparisons remain suggestive but not conclusive.

Action items.

- **[writing]** The paper presents a novel benchmark for scientific lineage reasoning, but the evidentiary strength of the experimental results is currently insufficient to support the specific claims of system differentiation and ranking stability. The primary concern is the complete absence of variance reporting in the main results. Table 1 presents exact accuracy scores for 14 different systems on the \geneexam{} benchmark (1,029 instances) as single point estimates (e.g., 27.3% for Claude Code vs 23.1% for

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_statistical_analysis.md`

The statistical reporting in the results section is generally clear but lacks necessary uncertainty quantification for the reported point estimates, leading to potential false precision.

First, **exact accuracy** in Table 1 and Section 5.2 is reported to one decimal place (e.g., 27.3% for the best system). With a total of 1,029 instances, the standard error for a proportion of 0.273 is approximately $\sqrt{0.273(1-0.273)/1029} \approx 0.014$ (1.4%). Reporting a value like 27.3% implies a precision of 0.1%, which is an order of magnitude finer than the statistical noise. The authors should round these to integers (e.g., 27%) or report them with a confidence interval (e.g., 27.3% ± 1.4%) to accurately reflect the stability of the metric.

Second, the **Population-Evolution Score (PES)** and its decomposition (Heredity, Variation, Selection) in Section 5.3 and Figure 5 are presented as precise means (e.g., "Heredity 61.9" vs "84.2"). These scores are derived from a panel of only 3 model judges. The text mentions inter-judge agreement (Krippendorff's $\alpha = 0.74$) but does not report the standard deviation or standard error of the mean scores across the judges or across the 30 tasks per system. Without this, the reader cannot determine if the observed gaps (e.g., the +5.4 Heredity gain) are statistically robust or within the noise of the judging process. The authors should report the standard deviation or 95% confidence intervals for the PES components.

Third, the **correlation claim** in Section 5.3 ("Spearman $\rho = 0.82$“) between ELO and PES rankings is made without a p-value or confidence interval. With a sample size of only 14 systems, the sampling distribution of the correlation coefficient is wide. A value of 0.82 is high, but without a significance test or CI, the claim of a "moderate correlation" (the text actually says "moderate" but 0.82 is strong) and the interpretation of the divergence lack statistical grounding. A p-value or bootstrap CI for ρ is required.

These are reporting fixes (writing) as the raw data (per-instance scores, per-judge scores) presumably exists in the logs mentioned in the appendix. No re-running of experiments is required, but the manuscript must be updated to reflect the uncertainty inherent in the sample sizes and judge counts.

Action items.

- [**writing**] Table 1 and Section 5.2 report exact accuracy to one decimal place (e.g., 27.3%) for 1,029 instances. This implies a precision of ~0.1%, but the standard error for a proportion p=0.273 with n=1029 is ~1.4%. Report accuracy as an integer or with a confidence interval (e.g., 27.3% ± 1.4%) to avoid false precision.
- [**writing**] Section 5.3 claims PES gains are 'Heredity-driven' and cites specific values (e.g., Heredity 61.9 vs 84.2) derived from a 3-judge panel. No uncertainty metric (SD, SE, or CI) is reported for these mean scores. Given the small number of judges (n=3) and the potential for inter-judge variance, report the standard deviation or 95% CI for the PES and its components to support the magnitude of the claimed gaps.
- [**writing**] Section 5.3 reports a Spearman correlation (= 0.82) between ELO and PES rankings without a p-value or confidence interval. With only 14 systems, this correlation is subject to high sampling variance. Report the 95% CI for the correlation coefficient or the p-value to confirm the statistical significance of the divergence claim.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_writing_quality.md`

The manuscript demonstrates a high standard of writing quality, allowing the reader to move through the complex theoretical framework and experimental results with minimal friction. The prose is precise, the sentence structures are generally clean, and the logical flow between sections is well-maintained.

The abstract effectively summarizes the benchmark's construction, the two evaluation modes (IG-Exam and IG-Arena), and the headline empirical findings without burying the main points. The introduction successfully establishes the "scientific lineage competence" gap and introduces the *Idea Genome* framework with clear, statable definitions.

Paragraphs are consistently well-structured, each focusing on a single point. For instance, the "Data Construction and Quality Assurance" section in Section 4 uses a numbered list to break down the four stages of construction, followed by a distinct paragraph for quality assurance metrics, preventing the common issue of mixing methodology with validation results in a single block. Transitions between the framework definition (Section 3) and the benchmark instantiation (Section 4) are smooth, with the text explicitly linking the abstract concepts to the concrete dataset.

Terminology is used consistently throughout; terms like "genome," "genomediff," and the specific task axes (T1–T4) are defined early and maintained without confusing synonym switching. The tense usage is appropriate, shifting logically between the present tense for describing the framework and the past tense for reporting experimental results.

While the text is dense with technical concepts, the authors avoid unnecessary wordiness. Sentences are generally direct, and complex ideas (such as the Population-Evolution Score decomposition) are broken down into clear, itemized components. There are no instances of garden-path sentences or ambiguous pronoun references that would force a re-read. The paper is a model of clear technical communication.